

Reflection-Boundary Equidistribution and Extensive Mahler Packet Pressure for a Hénon Horseshoe

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Abstract

For the area-preserving Hénon map $H(x, y) = (6 - x^2 - y, x)$, previous full-horseshoe and algebraic-exhaustion results make every odd primitive mixed-axis coordinate divisor reduced, totally real, and of degree $D_n = \sum_{d|n} \mu(n/d) 2^{(d+1)/2}$. Ordinary coordinate Weil height is uniformly bounded, so its fixed-parameter pressure is flat. We identify the correct extensive packet limit. Primitive coordinates sampled at their marked reflection axis converge to a non-invariant reflection-boundary Bernoulli measure, while time-averaging the same selected orbits converges to the invariant maximal-entropy Bernoulli measure. The marked limit implies convergence of normalized logarithmic Mahler measures and a nonconstant linear packet pressure

$$\lim_{\substack{n \rightarrow \infty \\ n \text{ odd}}} \frac{1}{n} \log \left(D_n e^{-sn D_n^{-1} \log M(\tilde{\Psi}_n)} \right) = \frac{1}{2} \log 2 - s\kappa_J,$$

where $0 < \kappa_J \leq \log(1 + \sqrt{7})$. Exact symbolic enumeration through period 21 and independent primitive-polynomial diagnostics through period 11 accompany the theorem. The result is a packet-level thermodynamic statement; it does not prove an individual height pressure, a prime trace, or a Hilbert–Pólya operator.

1 Introduction

Consider the scaled Hénon automorphism

$$H(x, y) = (6 - x^2 - y, x). \tag{1}$$

It is linearly conjugate to the repository normalization $(q, p) \mapsto (1 - 6q^2 - p, q)$. Arai’s certified hyperbolic plateau [1] transports the Devaney–Nitecki full-shift horseshoe [2] to parameter 6. Combining this with the Friedland–Milnor complex fixed-point count [3] exhausts every complex periodic point by the real full two-shift. In particular, the odd mixed-axis primitive coordinate polynomial is reduced, effective and totally real.

The integral recurrence

$$x_{j+1} = 6 - x_j^2 - x_{j-1} \tag{2}$$

gives the sharp periodic bound $|x_j| \leq 1 + \sqrt{7}$. Thus an ordinary fixed-parameter height weight is non-extensive. The first natural repair is the packet-average Mahler mass

$$a_n = \frac{1}{D_n} \log M(\tilde{\Psi}_n), \tag{3}$$

multiplied by the period clock n .

A subtlety appears before pressure: mixed-axis roots are marked at a reflection axis. They are sparse, symmetry-centered points rather than generic marked periodic points. We prove that their axis distribution is a reflected one-sided Bernoulli process, not the invariant two-sided Bernoulli measure. Time-averaging the same orbits does recover maximal entropy. This separation is the main structural result and prevents a false identification of the two Mahler slopes.

2 Equivariant full-shift coding

Define the involution

$$J(x, y) = (x, 6 - x^2 - y). \quad (4)$$

Then $J^2 = 1$ and $JHJ = H^{-1}$. Let σ be the left shift on $\{-, +\}^{\mathbb{Z}}$ and let

$$(\rho\omega)_k = \omega_{-k}. \quad (5)$$

Lemma 2.1 (Equivariant coding). *There is a homeomorphism $\pi : \{-, +\}^{\mathbb{Z}} \rightarrow \Lambda$, where Λ is the real chain recurrent set of (1), such that*

$$\pi\sigma = H\pi, \quad \pi\rho = J\pi. \quad (6)$$

Proof. At a Devaney–Nitecki large-parameter anchor, the sign itinerary is a one-to-one full-shift coding. Since J preserves the first coordinate and $H^k J = JH^{-k}$, that coding intertwines J with (5).

Continue the basic set and its conjugacy along Arai’s connected uniformly hyperbolic plateau from the anchor to parameter 6. Choose the structural continuation h_a in the standard small-neighborhood normalization. Both h_a and $J_a h_a J_{10}$ conjugate the anchor dynamics to the parameter- a dynamics and are close to the identity. Expansivity gives uniqueness in this normalization, so $J_a h_a J_{10} = h_a$. Composing the anchor itinerary with h_6 proves (6). $\square$

This is a specialization of standard symbolic/hyperbolic continuation machinery; Lind and Marcus [5] gives symbolic background, and Kang [4] gives broader reversible-periodic context. Neither source is used to assert the counting or limit theorem below.

3 Reflection half-words and primitive removal

Fix an odd integer $n = 2m + 1$. The intersection $\text{Fix}(\sigma^n) \cap \text{Fix}(\rho)$ consists of the words

$$(\omega_0, \omega_1, \dots, \omega_m, \omega_m, \dots, \omega_1), \quad (7)$$

so it has cardinality 2^{m+1} . Under lemma 2.1, these are exactly the marked points in $\text{Fix}(H^n) \cap \text{Fix}(J)$. The mixed-axis closure has degree 2^{m+1} and consists of simple points, so no symbolic point or algebraic root is missing.

Let Ω_n^{prim} denote the exact-period subset and put

$$D_n = \#\Omega_n^{\text{prim}} = \sum_{d|n} \mu(n/d) 2^{(d+1)/2}. \quad (8)$$

Lemma 3.1 (Primitive subtraction). *If $U_n = \text{Fix}(\sigma^n) \cap \text{Fix}(\rho)$, then*

$$\frac{\#(U_n \setminus \Omega_n^{\text{prim}})}{\#U_n} \leq \tau(n) 2^{-n/3}. \quad (9)$$

Consequently uniform measure on U_n and on Ω_n^{prim} differ in total variation by a quantity tending exponentially to zero.

Proof. Every proper divisor d of an odd integer n satisfies $d \leq n/3$. Bounding all lower-period contributions by $\sum_{d|n, d < n} 2^{(d+1)/2}$ and dividing by $2^{(n+1)/2}$ gives (9). The total variation identity follows because the primitive family is a subset of the full family. $\square$

In particular,

$$\lim_{\substack{n \rightarrow \infty \\ n \text{ odd}}} \frac{1}{n} \log D_n = \frac{1}{2} \log 2. \quad (10)$$

4 Two different weak-star limits

Let η_n be uniform measure on Ω_n^{prim} , with the reflection axis marked at coordinate zero. Let $(\xi_k)_{k \geq 0}$ be iid fair signs and form a two-sided sequence by

$$\omega_k = \omega_{-k} = \xi_k \quad (k \geq 0). \quad (11)$$

Denote its law by η_J .

Theorem 4.1 (Reflection-boundary limit). *As odd n tends to infinity,*

$$\eta_n \xrightarrow{*} \eta_J. \quad (12)$$

The limit is not shift invariant and is not the fair two-sided Bernoulli measure μ_B .

Proof. For a fixed cylinder radius r and $n > 2r$, the free coordinates $\omega_0, \dots, \omega_r$ in (7) are independent fair signs. The negative coordinates are their reflected copies. This proves cylinder convergence for the full family; lemma 3.1 transfers it to the primitive family.

Under η_J , the event $\{\omega_{-1} = \omega_1\}$ has probability one, whereas it has μ_B -probability $1/2$. After applying one shift, the same event pulls back to $\{\omega_0 = \omega_2\}$, which has η_J -probability $1/2$. Hence η_J is not shift invariant. $\square$

Now erase the distinguished axis by averaging over each cyclic orbit:

$$\bar{\eta}_n = \frac{1}{n} \sum_{j=0}^{n-1} (\sigma^j)_* \eta_n. \quad (13)$$

Theorem 4.2 (Orbit-averaged limit). *For odd $n \rightarrow \infty$,*

$$\bar{\eta}_n \xrightarrow{*} \mu_B. \quad (14)$$

More precisely, for a cylinder C depending on coordinates $[-r, r]$,

$$|\bar{\eta}_n(C) - \mu_B(C)| \leq \frac{4r+1}{n} + \tau(n)2^{-n/3}. \quad (15)$$

Proof. First use all palindromes. A window centered at j contains two positions paired by reflection only if $2j \equiv -(a+b) \pmod{n}$ for some $a, b \in [-r, r]$. Because n is odd, each value of $a+b \in [-2r, 2r]$ produces at most one bad center, so there are at most $4r+1$. At every other center the window reads distinct free half-word coordinates and has exactly the fair Bernoulli law. Average over centers and then apply lemma 3.1. $\square$

5 Extensive packet pressure

Write $x(z)$ for the first scaled coordinate and set

$$\phi(z) = \log^+ |x(z)|. \quad (16)$$

This is continuous on the compact horseshoe and satisfies $0 \leq \phi \leq \log(1 + \sqrt{7})$. Define

$$\kappa_J = \int \phi d(\pi_* \eta_J), \quad \kappa_{\max} = \int \phi d(\pi_* \mu_B). \quad (17)$$

Proposition 5.1. *One has*

$$0 < \kappa_J \leq \log(1 + \sqrt{7}). \quad (18)$$

Proof. The upper bound is immediate. The constant negative itinerary represents the fixed point with first coordinate $-1 - \sqrt{7}$. By continuity, a finite cylinder around this sequence has $\phi > 0$. That cylinder has positive η_J -measure, proving strict positivity. $\square$

Let $\tilde{\Psi}_n \in \mathbb{Z}[T]$ be the monic integral primitive coordinate polynomial. Total reality and squarefreeness give

$$a_n := \frac{1}{D_n} \log M(\tilde{\Psi}_n) = \frac{1}{D_n} \sum_{\tilde{\Psi}_n(\alpha)=0} \log^+ |\alpha|. \quad (19)$$

The right side is the ϕ -average under $\pi_* \eta_n$.

Theorem 5.2 (Packet-Mahler pressure). *For every fixed $s \in \mathbb{R}$,*

$$\lim_{\substack{n \rightarrow \infty \\ n \text{ odd}}} \frac{1}{n} \log (D_n \exp(-sna_n)) = \frac{1}{2} \log 2 - s\kappa_J. \quad (20)$$

This pressure is nonconstant.

Proof. By theorem 4.1 and continuity, $a_n \rightarrow \kappa_J$. Combine this with (10). Nonconstancy follows from proposition 5.1. $\square$

Similarly, define

$$b_n = \frac{1}{nD_n} \sum_{\alpha} \sum_{j=0}^{n-1} \phi(H^j z_{\alpha}), \quad (21)$$

where $z_{\alpha} \in \text{Fix}(J)$ is the point with first coordinate α . Then theorem 4.2 gives $b_n \rightarrow \kappa_{\max}$ and

$$\lim \frac{1}{n} \log (D_n \exp(-snb_n)) = \frac{1}{2} \log 2 - s\kappa_{\max}. \quad (22)$$

6 Executable certificate

The primary certificate enumerates every primitive palindrome through odd period 21. For radius-two cylinders, the marked-axis total variation from fair Bernoulli stabilizes at $3/4$, whereas the orbit-averaged values decrease toward zero. It independently rebuilds the primitive integral polynomials and reports the diagnostics in table 1.

A second implementation uses a different least-period test through period 17 and the exact period-one/three polynomial sentinels. Normal and optimized test runs agree, and 26 contract mutations are rejected.

Table 1: Finite diagnostics. These decimals are not used to prove theorems 4.1, 4.2 and 5.2.

n	marked Mahler average	orbit average
1	0.895879734614027	0.895879734614027
3	0.693147180559945	0.767528364331349
5	0.592642696019558	0.775457104945456
7	0.648968099586674	0.786446733777541
9	0.660700003269176	0.790519108637561
11	0.670373466816018	0.789976524363231

7 Claim boundary and next theorem

The pressure in (20) weights the complete packet by its average Mahler mass. It is not the individual partition $\sum_{\alpha} e^{-snh(\alpha)}$. If $\tilde{\Psi}_n$ factors, different factors may have different Weil heights; weak convergence of all roots does not control that finer distribution.

The finite rows suggest $\kappa_J \neq \kappa_{\max}$, but a theorem requires certified cylinder enclosures and a tail estimate for (16). No prime-power family, von Mangoldt amplitude, determinant continuation, self-adjoint operator, zeta-zero correspondence, or Riemann-hypothesis statement is obtained here.

8 Conclusion

The reflection slice supports a genuine extensive packet pressure, but its natural marked measure is a boundary process rather than an invariant equilibrium state. Orbit averaging restores maximal entropy. This distinction turns a false shortcut into two exact limit theorems and isolates a concrete next task: rigorously separate the two Hénon Mahler slopes.

References

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